STAT 516 FINAL

50 pt total

1. (10pt) Let X have the triangular pdf

$$f(x) = \begin{cases} x & \text{if } 0 < x \leq 1; \\ 2 - x & \text{if } 1 < x \leq 2; \\ 0 & \text{otherwise.} \end{cases}$$

   (a) Verify that $f(x)$ is a pdf

   (b) Obtain the CDF of X and write it carefully

   (c) Find $P(0.3 < X \leq 1.5)$

   Solution key: $f(x)$ is positive and integrates up to 1; the cdf of X is as follows. $F(x) = 0$ if $x \leq 0$, $F(x) = \frac{x^2}{2}$ if $0 < x \leq 1$, $F(x) = 2x - \frac{x^2}{2} - 1$ if $1 < x \leq 2$ and $F(x) = 1$ if $x \geq 2$. $P(0.3 < X \leq 1.5) = P(X \leq 1.5) - P(X \leq 0.3) = 0.83$

2. (10pt) A fair coin is tossed $n = 2m$ times. What is the probability that the number of heads obtained will be an even number? You may use the fact that

$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = 2^{n-1}$$

Solution key: since the number of heads $X \sim b(2m, 0.5)$ we have

$$P(X=0)+P(X=2)+\cdots+P(X=2m) = \sum_{x=0}^{m} \binom{2m}{2x} / 2^{2m} = \frac{2^{2m-1}}{2^{2m}} = \frac{1}{2}$$

3. (10pt) Let X have the standard Cauchy density $f(x) = \frac{1}{\pi}\frac{1}{1+x^2}$, $-\infty < x < \infty$. Find the probability density function of $Y = g(X) = \frac{1}{X}$

Solution key: $g(X)$ is a strictly monotone function, there is $g^{-1}(y) = \frac{1}{y}$ and $g^{'}(x) = -\frac{1}{x^2}$. Thus,

$$f_Y(y) = \frac{f(1/y)}{|g'\left(\frac{1}{y}\right)|} = \frac{1/\pi \times 1/(1+1/y^2)}{1/(1/y^2)} = \frac{1}{\pi}\frac{1}{1+y^2}$$

and so $Y = \frac{1}{X}$ also has the standard Cauchy distribution

4. (10pt) A fair coin is tossed three times. Let X be the number of heads in three tosses and Y the absolute difference between the number of heads and the number of tails.

   (a) (4pt)Find the joint pmf of (X, Y)

   (b) (3pt)Find marginal pmf's of both X and Y. You can write down both joint and marginal pmf's in the same table.

   (c) (3pt) What are the expectations of X and Y?

Solution key:

(a) The table would look like this:

Y/X	0	1	2	3	P(Y=y)
1	0	$\frac{3}{8}$	$\frac{3}{8}$	0	$\frac{6}{8} = \frac{3}{4}$
3	$\frac{1}{8}$	0	0	$\frac{1}{8}$	$\frac{2}{8} = \frac{1}{4}$
P(X=x)	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1

(b) $\mathbb{E}X = 1.5 = \mathbb{E}Y$

5. (10pt)Let (X, Y) be jointly distributed with pdf $f(x, y) = 2$, for $0 < x < y < 1$, and 0 elsewhere.

   (a) (4pt)Find conditional distribution of X given Y by finding its pdf. What kind of distribution is it?

   (b) (4pt)Find conditional distribution of Y given X by, again, finding its pdf. What kind of distribution is it?

   (c) (2pt)Find $P\left(Y \geq \frac{1}{2}|X = \frac{1}{2}\right)$ and $P\left(X \geq \frac{1}{3}|Y = \frac{2}{3}\right)$

   Solution key:

   (a) The marginal distribution is $f_2(y) = \int_0^y 2\,dx = 2y$ when $0 < y < 1$ and 0 elsewhere. Therefore, the conditional distribution is $f(X|Y) = \frac{2}{2y} = \frac{1}{y}$, for $0 < x < y$ and so it is uniform on $(0, y)$.

   (b) first, find marginal distributions $f_1(x) = \int_x^1 2\,dy = 2 - 2x$ when $0 < x < 1$ and 0 elsewhere. Thus, the conditional distribution is $f(Y|X) = \frac{f(x,y)}{f_1(x)} = \frac{1}{1-x}$, for $x < y < 1$ and so it is uniform on $(x, 1)$

   (c) $P\left(Y \geq \frac{1}{2}|X = \frac{1}{2}\right) = \int_{1/2}^1 \frac{1}{1-x}\,dy = 1$ and $P\left(X \geq \frac{1}{3}|Y = \frac{2}{3}\right) = \int_{1/3}^{2/3} \frac{1}{y}\,dx = \frac{1}{2}$